An Empirical Analysis of Temporal Aggregation and Quality-Adaptive Thresholding for Degraded Surveillance Face Recognition

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***Abstract*—Surveillance face recognition must match probes degraded by blur, low resolution, noise, and compression against clean gallery images. Using frozen ArcFace ResNet-50 embeddings, we empirically study three retraining-free mechanisms: temporal aggregation, quality-weighted aggregation, and quality-adaptive decision thresholds. On the ChokePoint benchmark (17 known, 8 unknown identities from Portal 1) under seeded synthetic degradation across four severity levels, severe degradation raises single-frame Equal Error Rate (EER) from 0.8% to 8.5% and lowers TAR@FAR=1% from 100.0% to 70.2%. Simple uniform 5-frame mean pooling recovers roughly half of this gap (TAR@1% 83.1%, EER 4.6%). A Laplacian-variance quality heuristic is non-monotonic under sensor noise and shows no significant correlation with genuine match similarity ($r = -0.051$, $p = 0.07$), whereas embedding norm shows a moderate within-tier correlation (Spearman $r = 0.20$ to 0.33). On tracks with mixed per-frame quality, norm-guided best-frame selection clearly outperforms random-frame and Laplacian-guided selection (Rank-1 99.6% vs. 96.1% and 94.5%), but uniform pooling performs at least as well (Rank-1 100.0%), and we observed no measurable benefit of quality weighting over uniform pooling. When all frames in a track are equally degraded, sharper quality weighting reduces accuracy relative to uniform pooling. In a simulated ID-switch test (20% contamination) in which the intruder frame has a higher norm than the genuine frames, best-frame selection degrades sharply (ΔRank-1 $= -46.3$ pp) while uniform pooling remains stable (-0.8 pp). Finally, a quality-adaptive threshold (AOSM) gave no gain over a fixed threshold at matched false alarm rates (-4.7 pp DIR at FAR=1%). These results come from one small dataset with synthetic degradation and simulated tracks; confidence intervals and validation on native surveillance data remain future work.**

***Index Terms*—Face recognition, surveillance biometrics, temporal aggregation, quality estimation, open-set verification, ArcFace, empirical benchmark.**

I. Introduction

Deploying deep face recognition models in real-world surveillance environments remains an open operational challenge in computer vision. Deep feature extractors, such as ArcFace [1] and CosFace [2], achieve near-perfect verification accuracy on standard benchmarks (e.g., LFW, CFP-FP) when probe and gallery images reside in similar high-quality feature distributions. However, operational surveillance systems frequently match unconstrained, low-quality probe video tracks captured by Closed-Circuit Television (CCTV) cameras against clean, frontal reference photographs enrolled in an access-control or attendance gallery.

This asymmetry, referred to here as the *clean-vs-degraded quality gap*, involves compounded physical degradations: optical blur, aggressive spatial downsampling, sensor noise, and lossy compression [4]. When applied to such degraded probe frames, frozen deep feature extractors output displaced embedding vectors, substantially reducing true accept rates at strict operational false alarm rates (FAR $\leq 1\%$).

To address this challenge without the computational expense and risk of catastrophic forgetting associated with end-to-end retraining or domain adaptation, practitioners can consider post-hoc, retraining-free mechanisms:

1) **Temporal Aggregation:** Combining multi-frame embeddings along a tracked video trajectory (either via uniform averaging or quality-weighted pooling) to generate a consolidated tracklet representation.
2) **Quality-Adaptive Thresholding:** Dynamically modulating the match threshold or classification boundary based on estimated probe quality.

These mechanisms are easy to evaluate optimistically: tracklets may be assumed free of ID switches, quality weighting may be reported without a uniform-averaging baseline, and threshold adaptation may be reported at a single operating point rather than at matched false alarm rates. We therefore evaluate each mechanism against the simplest alternative under matched protocols.

In this work, we present a controlled empirical investigation into the behavior of frozen ArcFace representations under surveillance degradations on the ChokePoint benchmark. Specifically, our paper provides five empirical observations:

- **Baseline Quality Gap Measurement:** We quantify the degradation trajectory of ArcFace ResNet-50 across clean, mild, medium, and severe conditions under an evaluation protocol comprising 8,160 negative impostor comparisons, observing that severe degradation drops TAR@FAR=1% by 29.8 pp and raises EER by roughly an order of magnitude.
- **Noise-Inflated Quality Flaw:** We observe that spatial-domain blur metrics (such as Laplacian variance) fail under high-frequency additive sensor noise, scoring severely corrupted frames higher than medium-quality frames ($r = -0.051$, $p = 0.07$). In contrast, deep feature magnitude ($\|x\|_2$) drops under severe corruption and correlates positively with genuine similarity across and within tiers (nominal $p \leq 1.07 \times 10^{-4}$).
- **Aggregation Dynamics:** On mixed-quality tracks, norm-

guided best-frame selection outperforms random and Laplacian-guided selection, but uniform mean pooling matches or exceeds every quality-aware strategy in our tests. On tracks with pervasive severe degradation, uniform pooling clearly outperforms sharper weighting, which is consistent with (but does not prove) an ensemble noise-cancellation effect.

- **Quality-Adaptive Thresholding (AOSM):** Evaluated across swept thresholds at matched false alarm rates, quality-dependent additive score shifts gave no gain over a fixed threshold (DIR@FAR=1%: 91.8% vs. 96.5%); a likely explanation is that quality relaxation boosts low-quality impostors as much as low-quality genuine probes.
- **Simulated Tracker Contamination:** In a simulated ID-switch test (20% intruder frames), uniform mean pooling is resilient (ΔRank-1 $= -0.8$ pp), whereas best-frame and sharpened weighting degrade severely (ΔRank-1 $= -46.3$ pp) when the intruder frame has a higher norm than the genuine frames.

II. Related Work

A. Deep Face Recognition & Domain Gap

Deep metric learning using angular margin loss functions (ArcFace [1], CosFace [2]) has defined modern face recognition. These models map aligned 112×112 face crops onto a unit hypersphere where identity separation is maximized. However, when deployed in surveillance environments, sensor noise, downsampling, and blur displace probe vectors away from clean gallery centroids. Prior works have investigated feature-level domain adaptation using projection heads or adversarial alignment [5].

B. Video Face Aggregation & Quality Weighting

Video-based face recognition leverages temporal redundancy to compensate for frame-level corruption. Common aggregation paradigms include uniform temporal pooling, attention-based pooling, and quality-aware pooling [6], [7]. Wong et al. [3] investigated probabilistic patch-based image quality assessment for face selection and video recognition on ChokePoint. In quality-weighted aggregation, frame embeddings are weighted using estimated quality indices derived from spatial heuristics (e.g., Laplacian blur variance) or deep feature magnitudes (e.g., MagFace [8], AdaFace [9]).

Yang et al. [7] train a neural aggregation module that adaptively weights frame features and report gains over naive aggregation on large video face benchmarks, and Meng et al. [8] train a backbone whose feature magnitude is designed to reflect face quality. Both approaches require training. In contrast, we study what a frozen, standard ArcFace-trained backbone offers without additional training: whether its raw embedding norm carries quality information under synthetic surveillance degradation, whether quality-based weighting improves on uniform pooling, and how both behave under tracker ID switches. Our evaluation is much smaller in scale, so our results are complementary to, rather than directly comparable with, these works.

C. Open-Set Verification Protocols

Surveillance face recognition operates in an open-set regime where unregistered individuals outnumber enrolled subjects. Standard evaluation protocols measure False Accept Rate (FAR) versus True Accept Rate (TAR) or Detection and Identification Rate (DIR) [10]. A natural idea, which we also used in an earlier prototype, is to relax the decision boundary for low-quality inputs. Whether this helps depends on whether low-quality impostor probes are boosted as much as low-quality genuine probes, so we evaluate it at matched operating points.

III. Methodology & Problem Formulation

A. Feature Extraction and Problem Formulation

Let $I_{\text{gal}} \in \mathbb{R}^{H \times W \times 3}$ denote a clean reference frame enrolled for an identity $y \in \mathcal{Y}_{\text{known}}$. The enrolled gallery vector g_y is extracted using a frozen ArcFace backbone $f_\theta(\cdot)$:

$$g_y = \frac{f_\theta(I_{\text{gal}})}{\|f_\theta(I_{\text{gal}})\|_2} \in \mathbb{S}^{D-1}, \tag{1}$$

where $\mathbb{S}^{D-1}$ is the D-dimensional unit hypersphere ($D = 512$). During inference, a probe tracklet $\mathcal{T} = \{I_1, I_2, \ldots, I_N\}$ comprising N aligned face crops is captured. Each frame yields a raw feature vector $x_k = f_\theta(I_k)$, a normalized unit embedding $z_k = x_k / \|x_k\|_2$, and a scalar quality score q_k.

B. Quality Estimation Formulations

We analyze two contrasting quality estimation paradigms:

1) **Pixel-Domain Heuristic (Laplacian Variance):** Spatial sharpness is estimated via the variance of the discrete Laplacian operator $\nabla^2 I_k$ combined with image contrast (pixel standard deviation σ_k), with each component clipped prior to convex combination:

$$b_k = \min\left(\frac{\text{Var}(\nabla^2 I_k)}{500.0}, 1.0\right), \quad c_k = \min\left(\frac{\sigma_k}{80.0}, 1.0\right),$$
$$q_k^{\text{Lap}} = \text{clip}\left(0.6 \cdot b_k + 0.4 \cdot c_k, 10^{-4}, 1.0\right). \tag{2}$$

2) **Feature-Domain Magnitude (Embedding Norm):** Inspired by MagFace [8], the Euclidean magnitude before unit normalization $\|x_k\|_2$ is used as a quality score:

$$q_k^{\text{Norm}} = \text{clip}\left(\frac{\|x_k\|_2 - \nu_{\min}}{\nu_{\max} - \nu_{\min}}, 0.0, 1.0\right), \tag{3}$$

where $\nu_{\min} = 16.0$ and $\nu_{\max} = 22.0$ are fixed empirical bounds on the feature norms of the ArcFace ResNet-50 backbone used here. They affect only the AOSM quality score, since the softmax weights below are normalized within each track.

C. Tracklet Aggregation Continuum

To explore the continuum between uniform noise-cancellation and quality filtering, we use a temperature-scaled softmax aggregation scheme:

$$w_k(\beta) = \frac{\exp(\beta \cdot \tilde{q}_k)}{\sum_{j=1}^{N} \exp(\beta \cdot \tilde{q}_j)}, \quad \tilde{q}_k = \frac{q_k - \mu_q}{\sigma_q + \epsilon}, \tag{4}$$

where $\beta \geq 0$ controls the sharpness of the weighting distribution, and μ_q, σ_q are the mean and standard deviation of quality scores across the tracklet. The aggregated tracklet embedding $z_{\mathcal{T}}(\beta)$ is

$$z_{\mathcal{T}}(\beta) = \frac{\sum_{k=1}^{N} w_k(\beta) z_k}{\left\| \sum_{k=1}^{N} w_k(\beta) z_k \right\|_2}. \quad (5)$$

This spans a spectrum:

- $\beta = 0$: **Uniform Mean Pooling** ($w_k = 1/N$).
- $\beta \in (0, \infty)$: **Quality-Aware Softmax Aggregation**.
- $\beta \to \infty$: **Best-Frame Selection** ($w_{\arg\max q_k} = 1$, others 0).

D. Adaptive Open-Set Margin (AOSM)

In open-set surveillance, probe tracklet $z_{\mathcal{T}}$ is matched against the gallery matrix $G \in \mathbb{R}^{|\mathcal{Y}_{\text{known}}| \times D}$. The top-1 predicted gallery identity is $\hat{y} = \arg\max_y s_y$, where $s_y = g_y^\top z_{\mathcal{T}}$. Under a standard fixed threshold rule, the probe is accepted as known if $s_{\hat{y}} \geq \tau_{\text{base}}$. Under the Adaptive Open-Set Margin (AOSM), the acceptance threshold is relaxed for degraded inputs based on the norm-derived quality $q_{\mathcal{T}} = \frac{1}{N} \sum_k q_k^{\text{Norm}}$:

$$s_{\hat{y}} \geq \tau_{\text{adaptive}}(q_{\mathcal{T}}) = \max\left(\tau_{\text{base}} - (1 - q_{\mathcal{T}}) \cdot m, \tau_{\text{floor}}\right), \quad (6)$$

where $m = 0.08$ is the relaxation margin and $\tau_{\text{floor}} = 0.20$. When the floor is inactive ($\tau_{\text{adaptive}} > \tau_{\text{floor}}$), this condition is algebraically equivalent to an additive score shift:

$$s_{\hat{y}} + (1 - q_{\mathcal{T}}) \cdot m \geq \tau_{\text{base}}. \quad (7)$$

Hence, AOSM acts as an additive score boost proportional to probe degradation.

IV. Experimental Protocol

A. Dataset and Identity Splits

We evaluate on the ChokePoint video surveillance benchmark [3], which records subjects walking through access portals under multiple overhead camera viewpoints. We use Portal 1, comprising 25 identities, partitioned into:

- **Known Gallery Set:** 17 identities ($|\mathcal{Y}_{\text{known}}| = 17$). Each known identity is enrolled using a single clean reference face crop from sequence P1E (first available frontal crop).
- **Unknown Impostor Set:** 8 held-out identities ($|\mathcal{Y}_{\text{unknown}}| = 8$) used strictly to generate open-set impostor queries.

For each of the 17 known identities, non-overlapping probe tracklets of length $N = 5$ frames are extracted from subsequent frames in sequence P1E (15 tracks per identity, yielding 255 genuine tracks). For each of the 8 unknown identities, 30 probe frames are extracted, yielding 240 unknown probe queries.

B. Degradation Models

To simulate operational surveillance conditions, aligned 112×112 face crops are subjected to four degradation tiers rendered with OpenCV and PIL:

- **Clean:** Original resolution and contrast (112×112).
- **Mild:** Gaussian blur ($\sigma = 1.0$), JPEG compression ($Q = 75$).
- **Medium:** Gaussian blur ($\sigma = 2.0$), $2\times$ downsampling (bicubic), JPEG ($Q = 50$).
- **Severe:** Gaussian blur ($\sigma = 3.5$), $4\times$ downsampling to 28×28 (bicubic), additive zero-mean Gaussian sensor noise ($\sigma_n = 15$), JPEG ($Q = 25$).

All corruptions are generated with fixed pseudo-random seeds, so every method is evaluated on identical pixels. In addition, **mixed-quality tracks** are constructed by sampling the degradation level of each frame from {Clean, Mild, Medium, Severe} with probabilities $[0.2, 0.3, 0.3, 0.2]$.

C. Evaluation Metrics

We maintain two separate protocols:

1) **All-Pairs 1:1 Verification:** Every probe is evaluated against all 17 gallery identities. Let y^* denote the true identity. Genuine comparisons ($s_{\text{gen}} = g_{y^*}^\top z$) and impostor comparisons ($s_{\text{imp}} = g_j^\top z$, $j \neq y^*$, plus all unknown probes matched against all gallery entries) yield 255 genuine and 8,160 impostor comparisons. We report Rank-1 identification accuracy (%), True Accept Rate at fixed False Accept Rates (TAR@FAR=1%, TAR@FAR=0.1%), Equal Error Rate (EER, %), and ROC-AUC. TAR is read at the last operating point with FPR $\leq$ target FAR.
2) **Open-Set Identification (DIR@FAR):** A probe is correctly detected and identified only if the top-1 predicted identity $\hat{y}$ equals y^* and the top-1 similarity exceeds the threshold τ. The false alarm rate (FAR) is measured over unknown impostor probes only. We sweep $\tau_{\text{base}} \in [0.10, 0.85]$ and report DIR at matched operating points (FAR $= 1.0\%$ and 5.0%).

V. Experimental Results

A. Quantifying the Degradation Gap

Table I presents the baseline evaluation across degradation tiers. Under clean, mild, and medium conditions, ArcFace ResNet-50 maintains high verification performance (Rank-1 $\geq 99.6\%$, TAR@1% $\geq 96.5\%$). Under severe degradation, single-frame recognition declines substantially: Rank-1 drops by 13.7 pp, TAR@1% drops by 29.8 pp, and EER rises from 0.8% to 8.5%.

Uniform temporal mean pooling over a short window of $N = 5$ frames recovers approximately half of this gap: Rank-1 rises to 93.3% (+7.1 pp), TAR@1% rises to 83.1% (+12.9 pp), and EER is reduced nearly in half to 4.6%.

B. The Noise-Inflated Laplacian Failure

Table II illustrates the failure mode of spatial-frequency blur estimation under additive noise. From Clean to Medium, Laplacian variance declines as expected ($0.3887 \to 0.2154 \to 0.1746$). In the Severe tier, however, Laplacian quality rises to **0.3124**, nearly double that of Medium footage. High-frequency pixel variations introduced by additive Gaussian sensor noise inflate the second-order derivative $\nabla^2 I$, so heavily corrupted

TABLE I
EXPERIMENT 1: THE QUALITY GAP ACROSS DEGRADATION TIERS (FROZEN ARCFACE RESNET-50 ON CHOKEPOINT, 255 GENUINE TRACKS, 8,160 IMPOSTOR COMPARISONS).

Degradation Tier	Evaluation Mode	Rank-1 (%)	TAR@ FAR=1% (%)	TAR@ FAR=0.1% (%)	EER (%)	ROC-AUC
Clean	Single Frame	100.0	100.0	92.9	0.8	0.9996
	Mean Pooling ($N = 5$)	100.0	100.0	95.3	**0.4**	**0.9998**
Mild	Single Frame	100.0	99.2	90.6	1.1	0.9995
	Mean Pooling ($N = 5$)	100.0	100.0	92.9	**0.5**	**0.9997**
Medium	Single Frame	100.0	96.5	82.8	1.6	0.9988
	Mean Pooling ($N = 5$)	99.6	99.2	85.5	**0.8**	**0.9993**
Severe	Single Frame	86.3	70.2	40.0	8.5	0.9696
	Mean Pooling ($N = 5$)	**93.3**	**83.1**	**54.5**	**4.6**	**0.9902**
Severe gap closed	Mean Pooling vs. Single	+7.1 pp	+12.9 pp	+14.5 pp	−3.9 pp	+0.0206

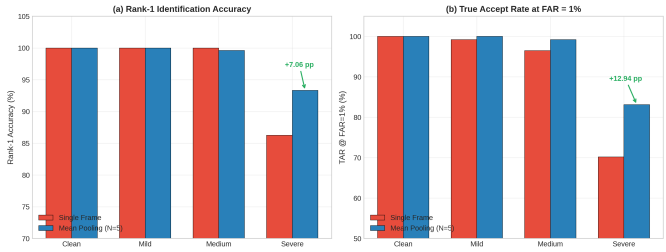


Fig. 1. Degradation gap across four tiers for a frozen ArcFace ResNet-50 backbone. Single-frame performance declines sharply under severe conditions (Rank-1: $100.0\% \rightarrow 86.3\%$; TAR@1%: $100.0\% \rightarrow 70.2\%$; EER: $0.8\% \rightarrow 8.5\%$), while uniform 5-frame mean pooling recovers approximately half the gap.

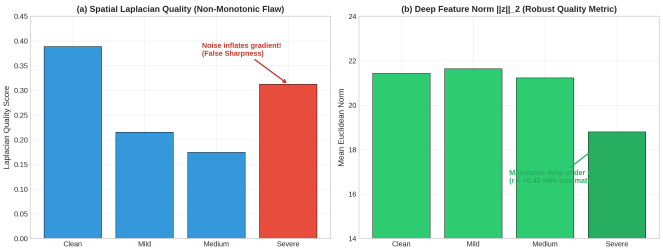


Fig. 2. Laplacian variance (pixel domain) vs. feature norm $\|x\|_2$ (feature domain) as quality proxies across degradation tiers. Laplacian variance rises non-monotonically in the Severe tier, consistent with sensor noise inflation ($r = -0.051$, $p = 0.07$), while the feature norm drops under severe noise and retains a positive correlation with genuine cosine similarity ($r = +0.4186$).

TABLE II
QUALITY METRIC PROGRESSION ACROSS DEGRADATION TIERS AND WITHIN-TIER SPEARMAN RANK CORRELATION BETWEEN FEATURE NORM AND GENUINE COSINE SIMILARITY (p-VALUES ARE NOMINAL AND ASSUME INDEPENDENT SAMPLES).

Degradation Tier	Mean q^{Lap}	Mean $\|x\|_2$	Spearman r ($\|x\|_2$)	Nominal p-value
Clean	0.3887	21.43	+0.2482	3.04×10^{-5}
Mild	0.2154	21.64	+0.2042	1.07×10^{-4}
Medium	0.1746	21.23	+0.2324	4.67×10^{-6}
Severe	**0.3124**	18.80	**+0.3281**	$\mathbf{4.83 \times 10^{-8}}$
Overall	Laplacian: $r = -0.0509$ ($p = 0.07$)		**+0.4186**	$\mathbf{2.92 \times 10^{-55}}$

TABLE III
SHARPNESS CONTINUUM ON ALL-SEVERE VIDEO TRACKS ($N = 5$ FRAMES PER TRACK): UNIFORM AVERAGING VS. SOFTMAX QUALITY WEIGHTING.

Weighting Scheme	Rank-1 (%)	TAR@1% (%)	TAR@0.1% (%)	EER (%)
Uniform (Mean Pooling)	**94.1**	**84.3**	**48.2**	**4.3**
Softmax ($\beta = 0.5$)	94.1	83.1	47.5	4.7
Softmax ($\beta = 1.0$)	92.9	80.4	46.7	5.2
Softmax ($\beta = 2.0$)	89.8	76.1	42.4	6.3
Softmax ($\beta = 5.0$)	87.1	72.2	36.5	7.9
Pure Best Frame ($\beta = \infty$)	87.5	67.8	36.1	9.0

frames appear sharp to the filter. Consistently, Laplacian quality shows no statistically significant correlation with genuine cosine similarity ($r = -0.0509$, $p = 0.07$).

In contrast, the feature norm $\|x\|_2$ remains relatively stable across the clean-to-medium tiers before dropping under severe corruption (18.80), and shows a positive rank correlation with genuine similarity overall ($r = +0.4186$) as well as within every individual tier ($r = +0.20$ to $+0.33$, nominal $p \leq 1.07 \times 10^{-4}$). A within-tier correlation of this size is moderate, not strong.

C. The Sharpness Continuum on Severe Video

In Table III, we test whether quality-weighted aggregation outperforms uniform averaging when all frames in a track are severely degraded. As the sharpness parameter β increases from $\beta = 0$ (uniform mean pooling) toward pure best-frame selection, performance trends downward: Rank-1 decreases from 94.1% to 87.5%, TAR@1% drops from 84.3% to 67.8%, and EER increases from 4.3% to 9.0%.

These results are consistent with the hypothesis that, when frames suffer from independent zero-mean sensor noise, temporal averaging provides an ensemble noise-cancellation effect that sharpened weights discard. Because quality is normalized within each track in Eq. (4), small frame-to-frame differences can also be amplified at high β, which offers an alternative explanation (see Section VII).

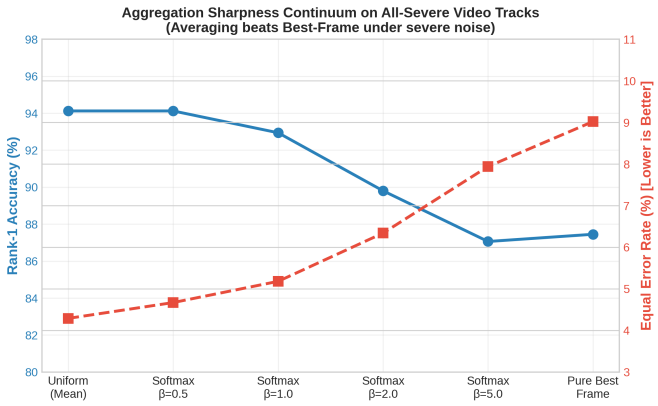


Fig. 3. Sharpness continuum on all-severe tracks ($N = 5$). As β increases from uniform pooling ($\beta = 0$) toward best-frame selection ($\beta \to \infty$), verification performance trends downward.

TABLE IV
AGGREGATION ON MIXED-QUALITY TRACKS (255 GENUINE TRACKS OF 5 FRAMES EACH, 240 UNKNOWN PROBES).

Strategy / Weighting	Rank-1 (%)	TAR@1% (%)	TAR@0.1% (%)	EER (%)
Worst Frame (Norm)	92.2	86.3	58.0	6.7
Random Single Frame	96.1	92.5	71.0	4.7
Best Frame (Laplacian)	94.5	88.6	64.7	6.0
Best Frame (Norm)	99.6	99.2	84.7	0.8
Uniform (Mean Pooling)	100.0	99.6	92.2	0.8
Softmax Norm ($\beta = 1.0$)	100.0	100.0	92.5	0.7
Softmax Norm ($\beta = 2.0$)	100.0	100.0	92.9	0.8

D. Aggregation Dynamics on Mixed-Quality Tracks

In Table IV, we evaluate aggregation strategies on tracks whose per-frame quality varies across the four tiers. Two observations follow:

- **The norm carries useful quality information.** Best-frame selection by norm reaches 99.6% Rank-1 and 0.8% EER, well above random single-frame capture (96.1% Rank-1, 4.7% EER) and Laplacian-guided selection (94.5% Rank-1, 6.0% EER), while selecting the lowest-norm frame is worst (92.2% Rank-1).
- **Uniform pooling is at least as good as every quality-aware strategy.** Uniform pooling and softmax weighting are at ceiling (100.0% Rank-1). The differences between them are one or two tracks out of 255 at TAR@1% and TAR@0.1%, which is within noise, and best-frame selection is lower than uniform pooling at TAR@0.1% (84.7% vs. 92.2%).

We therefore observed no measurable benefit of quality weighting over uniform pooling on this benchmark. Because the benchmark is close to ceiling, a harder mixture of tiers might separate the methods; we leave this to future work.

E. Simulated Tracker Contamination Test

In Table V, we evaluate the vulnerability of aggregation methods to tracking ID-switch errors by simulating 20% contamination: one frame out of five in a mixed-quality track

TABLE V
SIMULATED TRACKER CONTAMINATION ON MIXED TRACKS: 20% INTRUDER CONTAMINATION (ONE MILDLY DEGRADED INTRUDER FRAME PER 5-FRAME TRACK).

Contamination	Method	Rank-1 (%)	TAR@1% (%)	EER (%)
0% (Uncontaminated)	Uniform (Mean Pooling)	100.0	99.6	0.8
	Best Frame (Norm)	99.6	99.2	0.8
	Softmax ($\beta = 2.0$)	100.0	100.0	0.8
20% (1 Intruder Frame)	Uniform (Mean Pooling)	**99.2**	**97.3**	**1.2**
	Best Frame (Norm)	53.3	34.1	32.3
	Softmax ($\beta = 2.0$)	55.7	34.5	18.1
ΔRank-1 (Uniform vs. Best Frame)		−0.8 pp (Uniform)	vs. −46.3 pp (Best Frame)	

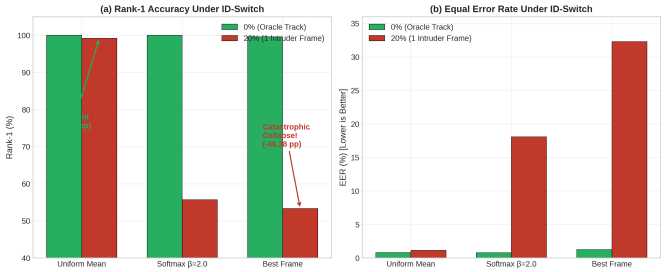


Fig. 4. Simulated ID-switch test with 20% intruder contamination (1 of 5 frames). Uniform mean pooling remains stable (ΔRank-1 $= -0.8$ pp), whereas best-frame and sharpened weighting degrade (ΔRank-1 $= -46.3$ pp) when the intruder frame has a higher feature norm than the genuine frames.

is replaced by a frame of a different identity. The intruder frame is sampled from the mildly degraded condition (norm ≈ 21.6), which often exceeds the norm of more degraded genuine frames in the track. This is a simulation of an ID switch, not a run of a real tracker.

- **Uniform mean pooling is resilient.** Rank-1 drops by only 0.8 pp (100.0% → 99.2%), and TAR@1% decreases from 99.6% to 97.3%. The four genuine frames retain majority influence on the pooled representation.
- **Quality-based selection is vulnerable in this setting.** Because the intruder frame has a higher norm than many genuine frames, best-frame selection often gives it full weight, and Rank-1 drops to 53.3% (−46.3 pp) with EER rising to 32.3%.

We did not run a control in which the intruder frame is drawn from the same quality distribution as the genuine frames, so the size of the drop reflects this design; the qualitative vulnerability of norm-based selection to a high-norm intruder does not depend on it.

F. Evaluation of Quality-Adaptive Thresholding (AOSM)

We evaluated open-set identification (DIR) across swept base thresholds, comparing a fixed threshold rule against AOSM on 255 mixed-quality known tracks and 240 mixed-quality unknown probe queries. The feature-norm quality q_T^{Norm} was used with margin $m = 0.08$ and floor $\tau_{\text{floor}} = 0.20$. At matched false alarm rates:

- **FAR = 1.0%:** fixed threshold DIR $= \mathbf{96.5}\%$, AOSM DIR $= \mathbf{91.8}\%$ (-4.7 pp).
- **FAR = 5.0%:** fixed threshold DIR $= \mathbf{99.2}\%$, AOSM DIR $= \mathbf{97.7}\%$ (-1.6 pp).

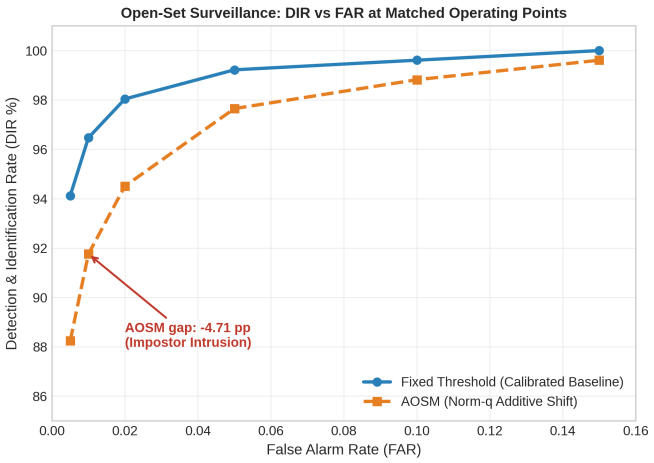


Fig. 5. Detection and Identification Rate (DIR) vs. False Alarm Rate (FAR) for a fixed threshold and for AOSM. At matched FAR $= 1\%$, AOSM reaches 91.8% DIR vs. 96.5% for the fixed threshold (-4.7 pp).

A likely explanation is that relaxing the acceptance threshold for low-quality probes also boosts low-quality impostor probes, which pushes the matched operating threshold upward. We tested a single margin value, so this result applies to that setting.

VI. Discussion & Practical Implications

Our observations suggest three cautious takeaways for deployments that use frozen face recognition backbones:

1) **Uniform pooling is a robust default.** Averaging embeddings over short tracklets ($N = 5$) recovers roughly half of the degradation gap at $O(N)$ cost. Norm-based quality selection clearly beats random and Laplacian-based selection, but we found no measurable benefit of quality weighting over uniform pooling, and uniform pooling was much more resilient to the simulated ID switches and to pervasive noise.
2) **Be cautious with spatial blur metrics under noise.** Laplacian variance is susceptible to high-frequency sensor noise. The embedding norm was a better quality signal than Laplacian variance in our synthetic setting, although its within-tier correlation with genuine similarity is only moderate.
3) **Prefer a calibrated fixed threshold.** Additive quality-based score relaxation did not improve performance at matched false alarm rates in our tests. Practitioners should calibrate a global threshold on an operational validation set.

VII. Limitations

The scope of this study is limited in the following ways:

1) **Gallery source and sequence overlap.** Gallery images are clean frontal frames from ChokePoint sequence P1E rather than identity documents, so we study a clean-vs-degraded quality gap, not a full cross-domain enrollment-to-surveillance gap. Probes come from the same sequence as the gallery frames, so clean-tier performance is likely optimistic; a cross-session split is needed.
2) **Synthetic degradation and independent noise.** Degradations were applied synthetically with noise drawn independently per frame. Independent zero-mean noise favors temporal averaging; real surveillance degradation is often correlated across consecutive frames.
3) **Within-track normalization.** Quality scores are normalized within each track (Eq. (4)), so when all frames are equally degraded, small score differences are amplified at high β. An absolute-scale weighting such as $w_k \propto \exp(\beta q_k^{\text{Norm}})$ with fixed global bounds should be investigated.
4) **AOSM scope.** AOSM was evaluated with a single margin ($m = 0.08$) and the feature-norm quality; other margins or quality sources may behave differently.
5) **Contamination design.** The ID-switch test used one intruder condition (a mildly degraded, higher-norm intruder frame) and simulated switches rather than a real tracker.
6) **Statistical power.** We report no confidence intervals; with 255 genuine tracks from 17 identities, one track corresponds to 0.39 pp, and many differences between methods amount to a few tracks. The mixed-quality benchmark is close to ceiling, and p-values are nominal. TAR@FAR=0.1% rests on about eight impostor comparisons.
7) **Pipeline and data scope.** We evaluate aligned face crops from 25 identities of one portal and one backbone. Detector failures, landmark misalignment, and tracker latency are not modeled. Validation on native surveillance data, such as QMUL-SurvFace [4], is needed to establish generality.

VIII. Ethics & Reproducibility

This work uses the publicly available ChokePoint research benchmark [3] under its academic terms of use. Surveillance face recognition carries dual-use risks and civil-liberties considerations; automated match scores should serve only as decision support with mandatory human review. Code, degradation configurations, and random seeds will be made publicly available.

IX. Conclusion

This paper presented an empirical evaluation of post-hoc mechanisms for surveillance face recognition with frozen ArcFace representations. On ChokePoint, uniform temporal pooling recovered roughly half of the synthetic quality gap. Embedding norm outperformed Laplacian variance as a quality signal in our synthetic setting, yet quality weighting gave no measurable benefit over uniform pooling and was vulnerable to simulated ID switches. Quality-dependent additive score shifts yielded no benefit at matched false alarm rates. These findings favor simple temporal pooling and call for caution with quality-based weighting and thresholding, pending confidence intervals and validation on native surveillance data.

REFERENCES

[1] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 4690–4699.

[2] H. Wang, Y. Wang, Z. Zhou, X. Ji, D. Gong, J. Zhou, Z. Li, and W. Liu, "CosFace: Large margin cosine loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 5265–5274.

[3] Y. Wong, S. Chen, S. Mau, C. Sanderson, and B. C. Lovell, "Patch-based probabilistic image quality assessment for face selection and improved video-based face recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2011, pp. 74–81, doi: 10.1109/CVPRW.2011.5981881.

[4] Z. Cheng, X. Zhu, and S. Gong, "Surveillance face recognition challenge," *arXiv preprint arXiv:1804.09691*, 2018.

[5] K. Sohn, S. Liu, G. Frostig, and M. Chandraker, "Unsupervised domain adaptation for face recognition in unlabeled videos," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 3210–3218.

[6] Y. Rao, J. Lin, J. Lu, and J. Zhou, "Learning discriminative aggregation network for video-based face recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 3781–3790.

[7] J. Yang, P. Ren, D. Zhang, D. Chen, F. Wen, H. Li, and G. Hua, "Neural aggregation network for video face recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 4362–4371.

[8] Q. Meng, S. Zhao, Z. Huang, and F. Zhou, "MagFace: A universal representation for face recognition and quality assessment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 14225–14234.

[9] M. Kim, A. K. Jain, and X. Liu, "AdaFace: Quality adaptive margin for face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 18750–18759.

[10] P. J. Phillips, P. J. Grother, and R. J. Micheals, "Evaluation methods in face recognition," in *Handbook of Face Recognition*, S. Z. Li and A. K. Jain, Eds. London: Springer, 2011, pp. 551–574.